Sunny Zhang

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EDUCATION

University of Waterloo, Faculty of Engineering September 2026 – April 2031 (Expected)
Candidate for Bachelor of Applied Science in Mechatronics Engineering

- **Awards:** Guo Family Engineering Scholarship ($5,000), President's Scholarship of Distinction ($2,000)

SKILLS

Languages: C++, C, Python, MicroPython
Embedded: STM32 (Mbed OS), Arduino, Raspberry Pi Pico, PlatformIO, CAN, I^2C, UART, PWM
Tools: Git/GitHub, CMake, Linux (WSL), VS Code, Fritzing
CAD: SolidWorks, Onshape, AutoCAD

EXPERIENCE

Mechanical Engineering Intern – Ace Manufacturing Inc. June 2026 – September 2026

- Produced **10+** design revisions in **AutoCAD** to customer requirements for a burn-off oven facility addition
- Designed **400+ litre** pump enclosures with covers, 4 m^2 platforms with stairs, and 3 m^2 interlocking steel panel walls to fit existing infrastructure, verifying structural safety with **shear stress calculations**
- Fabricated **20+** metal components and pipes for wastewater treatment systems on the lathe, mill, and bandsaw

DESIGN TEAM

Firmware Team Member – UW Robotics Team (Mars Rover) October 2026 – Present

- Completed the firmware onboarding challenge in **C++** on an **STM32F446RE** (**Mbed OS**), writing a reusable servo driver class that maps angle commands to 1–2 ms **PWM** pulses with configurable range and const-correct interfaces
- Built a **CAN bus** receiver app that sets servo position from float commands, and integrated it and the driver library into the team's **CMake** build
- Submitted the work for code review through a **GitHub** fork and pull request

PROJECTS

2D Mapping Vehicle September 2026 – Present

- Built an **Arduino**-based differential-drive robot that maps its surroundings in 2D using a servo-swept **time-of-flight sensor**, **quadrature encoders**, an **MPU6050 IMU**, and a multi-layered **SolidWorks** chassis
- Wrote **C++** firmware in **PlatformIO** that fuses encoder odometry with gyro heading to compute differential-drive kinematics for real-time position tracking
- Developed a live **Python** tool that parses **UART** serial data and plots the mapped environment and detected boundaries in real time
- Drafted **Fritzing** wiring diagrams for the dual-rail power system (L298N motor driver, buck-regulated 5 V logic)

Autonomous Robot Vacuum April 2026 – May 2026

- Designed an autonomous vacuum robot using a **Raspberry Pi Pico W**, **HC-SR04 ultrasonic sensors**, and an **MPU6050** gyroscope for obstacle sensing and heading control
- Programmed closed-loop **MicroPython** control for obstacle avoidance and course correction
- Iterated a 3D-printed **Onshape** chassis to protect the electronics from collisions
- Tuned avoidance logic to stop and reverse within **3 cm** of obstacles, maximizing floor coverage

AI Dashcam Prototype – SHAD Canada June 2025 – July 2025

- Collaborated in an **8-person** multidisciplinary team to define the system architecture and business case for an AI dashcam that automatically detects and reports road damage
- Co-presented the concept and market viability to industry judges, winning **'Best Presentation'** and **'Best Prototype'** out of **10 teams**